

Research on Motion Control of Wheel-legged Robot Based on ASMC and LQR

Yong Ding

Department of Automation
University of Science and
Technology of China Hefei,
China

yding1111@mail.ustc.edu.cn

Yong Wang*

Department of Automation
University of Science and
Technology of China
Hefei, China

yongwang@ustc.edu.cn

*Corresponding author

Yi Wang

Department of Automation
University of Science and
Technology of China
Hefei, China

wangyi0114@mail.ustc.edu.cn

Ke Wu

Department of Automation
University of Science and
Technology of China
Hefei, China

wkwwk@mail.ustc.edu.cn

Abstract—Wheel-legged robots combine the high-speed movement of wheeled robots with the superior obstacle-navigation abilities of legged robots. To enhance the robot's resilience against disturbances and improve its ability to overcome obstacles, we developed a control scheme that decouples the control into two parts: posture control and balance control. First, the robot's dynamics were modeled using the Euler-Lagrange method. To manage external disturbances, an Adaptive Sliding Mode Controller (ASMC) was designed to handle posture control. For balance control, a Linear Quadratic Regulator (LQR) was implemented to solve the Riccati equation in real-time, generating the necessary control torques to stabilize the robot. By combining these two control strategies, we present a framework that significantly improves the robot's mobility. Furthermore, tests conducted on a physical prototype demonstrated the effectiveness of the algorithm and the robot's disturbance rejection capability. The results confirm that the control scheme, which integrates ASMC and LQR, allows the wheel-legged robot to execute complex maneuvers and traverse uneven terrain with ease.

Index Terms—wheel-legged robot, LQR, ASMC;

I. INTRODUCTION

As the frontier field of modern robotics technology and research, the design of two-wheeled legged robots integrates the advantages of both wheeled and legged robots, demonstrating unique application potential and research value [1]. On flat or structured environments, the efficiency of wheeled movement provides the robot with the ability to move quickly; while in complex terrains or obstacle environments, the flexibility of the legged structure gives the robot superior adaptability and obstacle-crossing ability [2], [3]. This combination not only expands the robot's operating range but also greatly enhances its stability and maneuverability in changing environments [4]. Stability in two-wheeled legged robot control is the focus of research, and the SR600 robot developed by C. Zhang et al. uses PID to achieve robot balance control [5], based on which the SR600-II robot, capable of switching between wheeled and legged drive, was developed to enhance adaptability to complex terrains [6]. Xujiong Feng and his team proposed a new two-wheeled leg robot with a seven-link leg structure and designed LQR and active disturbance rejection control (ADRC) controllers to achieve control, which can make the robot pass through bumpy roads [7]. Victor Kjem and his team at ETH

Zurich have developed a two-wheeled legged robot called Ascento, which has only four motors all over its body and uses LQR and a heuristic feedforward control strategy. It can not only jump and rotate in place, but also control the height change of the robot, and can realize automatic recovery of balance after dumping [8], [9]. After that, the team used RL algorithm to achieve blind stair climbing [10]. Tencent's Ollie robot utilizes a parallel leg structure, which offers superior load-bearing capacity compared to a serial configuration. The robot is also equipped with a retractable tail mechanism, with a wheel mounted at the end. This allows the robot to maintain stable three-point support during high-speed movement on flat surfaces. Additionally, by swinging its tail, the Ollie robot can achieve longer jumps and perform complex maneuvers like backflips [11], [12].

In this paper, Wheel-legged robots are inherently unstable systems. Ensuring balance control while conducting posture control presents a significant challenge. During the robot's balanced motion, external disturbances can significantly impact its stability. The robot must adapt its posture to mitigate the effects of these disturbances. To address this issue, we have designed an adaptive sliding mode controller (ASMC) to tackle this problem, coupled with a Linear Quadratic Regulator (LQR) as the balance controller. This configuration endows the robot with enhanced disturbance rejection capabilities and a stable body posture. The implementation of LQR for balance control in robots has been realized. Experiments have been conducted on actual robots to validate the feasibility of the control scheme.

The main contributions of this paper are as follows:

1. An ASMC was designed for the Wheel-legged robots to handle external disturbances, and a LQR controller was designed to achieve balance control.
2. Verification of the feasibility and effectiveness of the control method.

II. DYNAMIC MODELLING AND CONTROLLER

We have developed a wheel-legged robot platform that weighs 16.2kg and has a maximum height of 0.6m, as shown in Fig.1. STM32 is used as the core processing unit, IMU

is used to obtain the robot attitude, SBUS remote control receiver receives the command required by the operator, and communicates with the motor through the CAN network to read the motor Angle, speed, current and send the control command. The robot consists of a total of six motors, with each leg consisting of three motors: a hip motor, a knee motor and a hub motor. The knee motor and the knee bearing are connected through a belt. Hub motors are direct drive motors that provide fast response speeds to keep the robot balanced. The leg structure consists of three motors that support the movement of the leg in a vertical plane, and the position of the foot can be adjusted by controlling the Angle of the knee (θ_{knee}) and the Angle of the hip (θ_{hip})

$$\begin{cases} x = L_{upper} \sin \theta_{hip} + L_{lower} \sin(\theta_{hip} + \theta_{knee}) \\ y = L_{upper} \cos \theta_{hip} + L_{lower} \cos(\theta_{hip} + \theta_{knee}) \end{cases} \quad (1)$$

Where L_{upper} is the length of the thigh and L_{lower} is the length of the calf.



Fig. 1. our Wheel-legged Robot

The motion control of the robot is divided into balance control and attitude control. Firstly, the dynamics model of the two-wheeled legged robot is constructed by Euler-Lagrange method. On the one hand, the state variables required for balance control are determined, the linear system model of the system is sorted out, and the optimization function for maintaining the balance and stability of the system is defined. The gain matrix is obtained by solving the Riccati equation to realize the balance control of the robot. On the other hand, the Lyapunov function required for attitude control is designed,

and the adaptive synovial controller is designed with adaptive law and synovial vector, and the stability of the control system is proved. Finally, experiments are carried out on the actual robot platform, and the experiments show that the designed two-wheeled legged robot has better stability and better load capacity.

A. Dynamic Modelling

When the robot leg length is fixed, the robot's dynamic model can be simplified to a basic two-wheel inverted pendulum model. Assuming no sliding between the robot wheels and the ground and that all structures are rigid bodies with uniform density. This paper uses the Lagrange method for robot modeling.

The equation for the kinetic and potential energy components is derived, and the Lagrange energy function is obtained.

$$L = N - P \quad (2)$$

where N represents the system's kinetic energy, and P represents the system's potential energy. By differentiating L , we get:

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{\theta}} - \frac{\partial L}{\partial \theta} = u \quad (3)$$

From this equation, the overall dynamic model is obtained as follows:

$$M(q)\ddot{q} + C(\dot{q}, q)\dot{q} + G(q) + f = u \quad (4)$$

where $M(q) \in \mathbb{R}^{n \times n}$ denotes the mass matrix, $C(\dot{q}, q) \in \mathbb{R}^n$ denotes the vector of Coriolis and centrifugal terms, $G(q) \in \mathbb{R}^n$ denotes the vector of gravity terms, $f \in \mathbb{R}^n$ represents the vector consisting of joint friction and external disturbances, and $u \in \mathbb{R}^n$ is the control input vector, which is related to the motor output torque by $\tau = J^T u$, where $J \in \mathbb{R}^{n \times n}$ is the Jacobian matrix of the robot, calculated based on the robot's kinematic model. Here is a more detailed modeling process [13].

B. Control System Design

To maintain the robot's balance and posture control, the control framework is designed as follows: a balance controller and a posture regulation controller are designed. The two controllers are responsible for different functions and work on different objects, so they can be designed independently [7]. The posture regulation controller is implemented through an adaptive sliding mode controller, which adjusts the robot's height and achieves coordinated control of the two legs, even realizing a walking gait. The balance control is implemented through the LQR controller, using feedback control from the robot's state data returned by the motors and IMU, with the control output to the hub motors to maintain balance. As shown in Fig.2, The diagram shows the overall implementation framework.

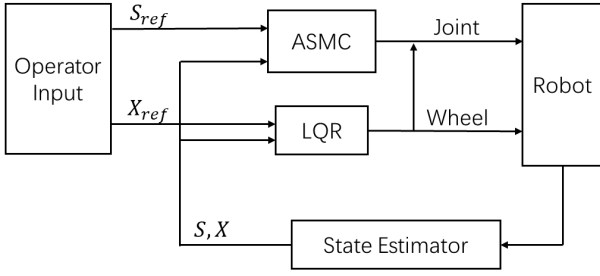


Fig. 2. System Control Block Diagram

C. Adaptive Sliding Mode Controller

From the actual robot system, it is known that the mass matrix M is a symmetric matrix and is bounded for any q : $\|M\| \leq \xi_M$. For gravity and friction terms, as well as unknown disturbances, they are also considered bounded with $\|G\| \leq \xi_G$, $\|f\| \leq \xi_f$. The Coriolis force C is also bounded and only depends on $\dot{q}$, that is, $\|C\| \leq \varsigma\|\dot{q}\|$. In addition, the matrix $\dot{M} - 2C$ is a symmetric matrix, that is, $Z^T(\dot{M} - 2C)Z = 0$.

For the modeling of robot dynamics systems, it is difficult to achieve accurate structure and parameters, but the controller depends on these models. For this purpose, an adaptive sliding mode controller is designed, with the adaptation law based on the upper bound of the uncertain parameters.

Design the sliding mode surface as $S = \dot{e} + \Lambda e$.

where $e = q_d - q$, represents the error vector, $S = [S_1, S_2, \dots, S_n]^T$ is the sliding surface vector, $\Lambda = \text{diag}(\Lambda_1, \Lambda_2, \dots, \Lambda_n)$ is a diagonal positive definite matrix.

For this system, in order to facilitate the design of the adaptive rate and the derivation of the control law, the Lyapunov function is designed as follows:

$$V(t) = \frac{1}{2}(S^T M S + \tilde{\rho}^T \Omega \tilde{\rho}) \quad (5)$$

where $\Omega = \text{diag}(\Omega_1, \Omega_2, \dots, \Omega_n)$ is a diagonal positive-definite matrix, and $\tilde{\rho} = \hat{\rho} - \rho$ denotes the estimation error vector of the upper bounds of uncertain parameters $\rho \in \mathbb{R}^n$.

Taking the derivative of $V(t)$ with respect to time yields

$$\dot{V}(t) = S^T M \dot{S} + S^T \left[\frac{1}{2}(\dot{M} - 2C) + C \right] S + \tilde{\rho}^T \Omega \dot{\tilde{\rho}} \quad (6)$$

Based on the dynamic properties of the robot, substituting (4) into (6) yields

$$\dot{V}(t) = S^T [u - (Cq + G + f) - M(\ddot{q}_d - \Lambda \dot{\tilde{q}}) + CS] + \tilde{\rho}^T \Omega \dot{\tilde{\rho}} \quad (7)$$

Define the trajectory $\ddot{q}_r = \ddot{q}_d - \Lambda \dot{\tilde{q}}$ to facilitate the subsequent derivation and stability proof of the controller.

Substituting $\ddot{q}_r$ into (7), we finally get:

$$\dot{V}(t) = S^T [u - (M\ddot{q}_r + Cq_r + G + f)] + \tilde{\rho}^T \Omega \dot{\tilde{\rho}} \quad (8)$$

Design the control law as:

$$u = \widehat{M}\ddot{x}_r + \widehat{C}\dot{x}_r + \widehat{G} + \widehat{f} - K_d \text{sgn}(S) \quad (9)$$

where $\widehat{M}, \widehat{C}, \widehat{G}, \widehat{f}$ are the estimates of the system dynamics parameters, and $K_d = \text{diag}(K_{d1}, K_{d2}, \dots, K_{dn})$ is a diagonal positive-definite matrix.

Substituting (9) into (8):

$$\begin{aligned} \dot{V}(t) = & S^T [(\widehat{M} - M)\ddot{q}_r + (\widehat{C} - C)\dot{q}_r + (\widehat{G} - G) \\ & + (\widehat{f} - f) - K_d \text{sgn}(S)] + \tilde{\rho}^T \Omega \dot{\tilde{\rho}} \end{aligned} \quad (10)$$

Given the boundedness of the robot model parameters with upper bounds ρ , the equation can be rewritten as:

$$\|\widetilde{M}\ddot{x} + \widetilde{C}\dot{x} + \widetilde{G} + \widetilde{f}\| \leq \sum_{i=1}^n \rho_i \quad (11)$$

Therefore, equation(10) is equivalent to

$$\dot{V}(t) \leq \sum_{i=1}^n |S_i|(\rho_i - K_{di}) + \tilde{\rho}^T \Omega \dot{\tilde{\rho}} \quad (12)$$

Define the positive-definite diagonal matrix $K_c = \text{diag}(K_1, K_2, \dots, K_n)$ such that:

$$K_d = K_c + \rho \quad (13)$$

then, substituting (14) into (13)

$$\dot{V}(t) \leq \sum_{i=1}^n |S_i|(-\tilde{\rho}_i - K_{di}) + \tilde{\rho}^T \Omega \dot{\tilde{\rho}} \quad (14)$$

Take the adaptation law:

$$\dot{\tilde{\rho}}_i = |S_i| \Omega_i^{-1} \quad (15)$$

Substitute (16) into (14), simplify $\dot{V}(t)$:

$$\dot{V}(t) \leq -K_i |S_i| \leq 0 \quad (16)$$

$\dot{V}(t)$ is semi-negative definite, which proves that the controller we designed and the adaptive law we selected can stabilize the system, and the system state trajectory eventually converges to the sliding surface $S = \dot{e} + \Lambda e = 0$.

D. LQR Controller

The LQR (Linear Quadratic Regulator) control is a control method based on linear quadratic optimization theory, used to design optimal controllers for linear time-invariant systems to meet certain performance criteria such as stability, speed, and accuracy. The core idea is to obtain the feedback gain matrix K by minimizing the optimization function, achieving system stability through feedback control $U = -KX$.

Based on the dynamic model, select the state variables $X = [\theta, \dot{\theta}, x, \dot{x}, \phi, \dot{\phi}]^T$ and $U = [T, T_P]^T$, establishing the nonlinear dynamic model as:

$$\dot{X} = f(X) + g(X)U \quad (17)$$

Considering the tracking of target states and energy consumption, define the optimization function as:

$$J = \int_0^\infty (X^T Q X + U^T R U) dt \quad (18)$$

where Q and R are weight matrices, both being diagonal positive-definite matrices. The size of each diagonal element represents the weight of the corresponding state component in the optimization process.

By solving the Riccati equation:

$$A^T P + P A - P B R^{-1} B^T P + Q = 0 \quad (19)$$

the P matrix can be obtained, and the gain matrix satisfies:

$$K = -R^{-1} B^T P \quad (20)$$

To make the robot track the reference trajectory, the control input satisfies:

$$U = -K(X - X_{\text{ref}}) \quad (21)$$

where the reference state can be $X_{\text{ref}} = [x_{\text{ref}} \ \dot{x}_{\text{ref}} \ 0 \ 0 \ 0 \ 0]^T$, depending on the specific task.

The above controller can maintain the robot's balance stability and achieve trajectory tracking.

III. EXPERIMENTS

Test the posture control and balance control on a self-built robot platform, conducting robot speed tracking experiments, pitch control experiments, and robot load height variation experiments.

1) *Speed Control*: In robot balance control, speed tracking is particularly important in maintaining balance, as it reflects the performance of the designed LQR balance controller.

As shown in Fig.3, the robot's speed response is fast, reaching the target reference speed and maintaining it closely, with a maximum speed of 1.75 m/s. During the tuning process of the speed tracking experiment, choosing larger weight coefficients allows the robot to track the target speed more quickly, but signs of instability may also occur.

2) *Posture Control*: Robot load experiment, testing the attitude control ability of the robot under different loads, and the adjustment ability of the adaptive parameter K . Under different loads, the ability of the robot to control leg length was tested. It can be seen that a load of 5kg has little effect on the adjustment of leg length in Fig.5. There is a certain steady-state error in height control, but the actual error is only 0.3cm, which will not affect the robot's balance control and can also meet our requirements for attitude control. At a load of 5kg(Fig.4), the steady-state error reached 0.4cm, but the

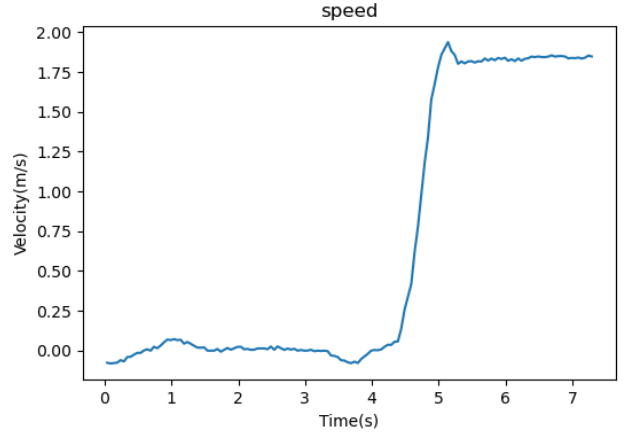


Fig. 3. Speed tracking experiment

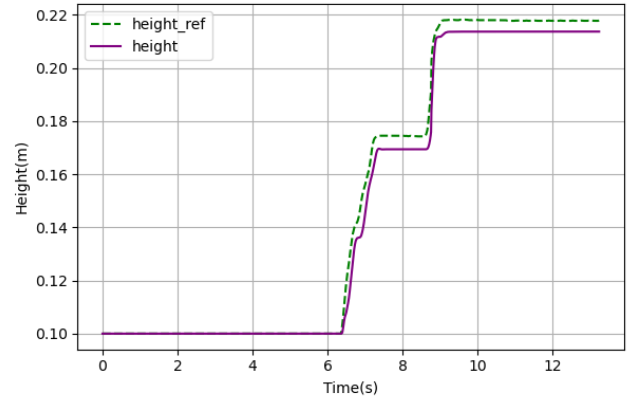


Fig. 4. Height variation experiment with a load of 5Kg

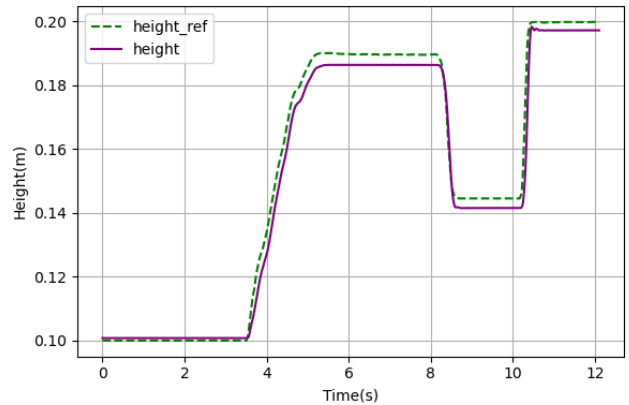


Fig. 5. Height variation experiment without load

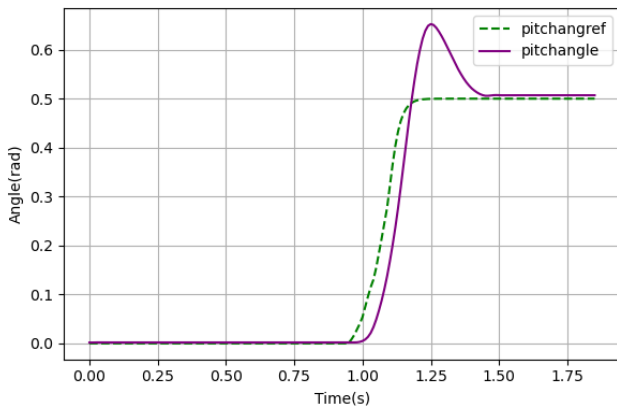


Fig. 6. Pitch tracking experiment

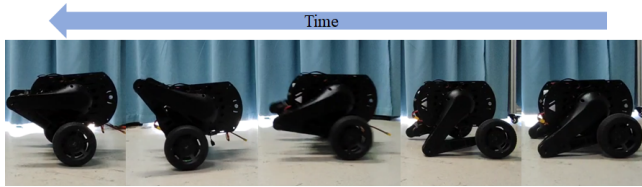


Fig. 7. Photos of the robot at different moments during the process of regaining balance

response speed was fast and almost unaffected, still meeting the requirements of attitude control.

The control experiment of the pitch angle of the robot was conducted, and the experimental results are shown in Fig.6. It can be seen from the figure that there is a certain overshoot in the pitch angle control, but there is almost no steady-state error, and the response speed is fast, which can fully meet the application requirements of practical robots.

As illustrated in the Fig.7, when the robot is in a falling state, it can still achieve balance control through our proposed control algorithm. This series of images depicts actual snapshots taken at various time intervals from right to left. Initially, the robot adjusts its leg length and subsequently moves swiftly backward to restore itself to a balanced state.

IV. CONCLUSION

This paper addresses the issues of posture control and balance control in a wheel-legged robot by designing and implementing ASMC and LQR. These controllers were deployed on a real wheel-legged robot, and experiments such as speed tracking and varying the center of mass under load conditions were conducted. The real-world experiments demonstrated that our mechanical structure design, system modeling, and control theory are correct and feasible. The designed ASMC and LQR controllers performed effectively. Even after increasing the degrees of freedom in the robot's legs, it was able to maintain balanced motion with varying center of mass heights. Additionally, the robot exhibited strong load-bearing capacity and the ability to flexibly adjust its posture. Moving forward, we plan to design a safety controller to provide safety assurance

for the robot, or integrate external sensors such as radar and cameras to enable autonomous navigation and other advanced functions.

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